Computer Networks Lab

Exp 3:Analysis of network Traffic using Network Monitoring tools

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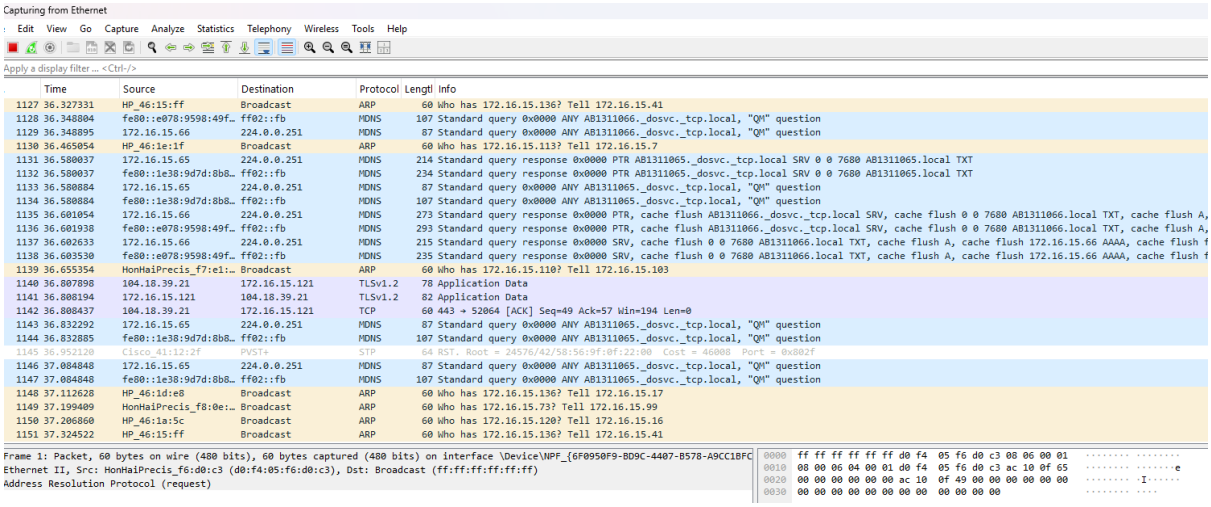
Capturing live network Traffic:

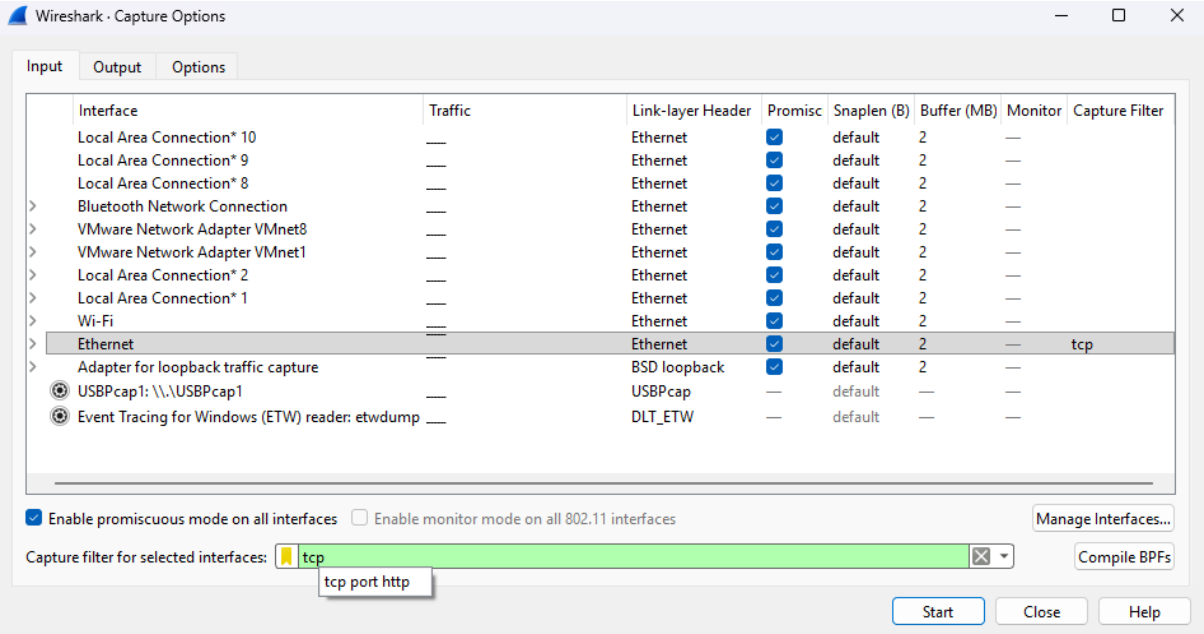


	Time	Source	Destination	Protocol	Length	Info
1127	36.327331	HP_46:15:ff	Broadcast	ARP	60	Who has 172.16.15.136? Tell 172.16.15.41
1128	36.348804	fe80::e078:9598:49f…	ff02::fb	MDNS	107	Standard query 0x0000 ANY AB1311066._dosvc._tcp.local, "QM" question
1129	36.348895	172.16.15.66	224.0.0.251	MDNS	87	Standard query 0x0000 ANY AB1311066._dosvc._tcp.local, "QM" question
1130	36.465054	HP_46:1e:1f	Broadcast	ARP	60	Who has 172.16.15.113? Tell 172.16.15.7
1131	36.580037	172.16.15.65	224.0.0.251	MDNS	214	Standard query response 0x0000 PTR AB1311065._dosvc._tcp.local SRV 0 0 7680 AB1311065.local TXT
1132	36.580037	fe80::1e38:9d7d:8b8…	ff02::fb	MDNS	234	Standard query response 0x0000 PTR AB1311065._dosvc._tcp.local SRV 0 0 7680 AB1311065.local TXT
1133	36.580884	172.16.15.65	224.0.0.251	MDNS	87	Standard query 0x0000 ANY AB1311065._dosvc._tcp.local, "QM" question
1134	36.580884	fe80::1e38:9d7d:8b8…	ff02::fb	MDNS	107	Standard query 0x0000 ANY AB1311065._dosvc._tcp.local, "QM" question
1135	36.601054	172.16.15.66	224.0.0.251	MDNS	273	Standard query response 0x0000 PTR, cache flush AB1311066._dosvc._tcp.local SRV, cache flush 0 0 7680 AB1311066.local TXT, cache flush A,
1136	36.601938	fe80::e078:9598:49f…	ff02::fb	MDNS	293	Standard query response 0x0000 PTR, cache flush AB1311066._dosvc._tcp.local SRV, cache flush 0 0 7680 AB1311066.local TXT, cache flush A,
1137	36.602633	172.16.15.66	224.0.0.251	MDNS	215	Standard query response 0x0000 SRV, cache flush 0 0 7680 AB1311066.local TXT, cache flush A, cache flush 172.16.15.66 AAAA, cache flush f
1138	36.603530	fe80::e078:9598:49f…	ff02::fb	MDNS	235	Standard query response 0x0000 SRV, cache flush 0 0 7680 AB1311066.local TXT, cache flush A, cache flush 172.16.15.66 AAAA, cache flush f
1139	36.655354	HonHaiPrecis_f7:e1:…	Broadcast	ARP	60	Who has 172.16.15.110? Tell 172.16.15.103
1140	36.807898	104.18.39.21	172.16.15.121	TLSv1.2	78	Application Data
1141	36.808194	172.16.15.121	104.18.39.21	TLSv1.2	82	Application Data
1142	36.808437	104.18.39.21	172.16.15.121	TCP	60	443 → 52064 [ACK] Seq=49 Ack=57 Win=194 Len=0
1143	36.832292	172.16.15.65	224.0.0.251	MDNS	87	Standard query 0x0000 ANY AB1311065._dosvc._tcp.local, "QM" question
1144	36.832885	fe80::1e38:9d7d:8b8…	ff02::fb	MDNS	107	Standard query 0x0000 ANY AB1311065._dosvc._tcp.local, "QM" question
1145	36.952120	Cisco_41:12:2f	PVST+	STP	64	RST. Root = 24576/42/58:56:9f:0f:22:00 Cost = 46008 Port = 0x802f
1146	37.084848	172.16.15.65	224.0.0.251	MDNS	87	Standard query 0x0000 ANY AB1311065._dosvc._tcp.local, "QM" question
1147	37.084848	fe80::1e38:9d7d:8b8…	ff02::fb	MDNS	107	Standard query 0x0000 ANY AB1311065._dosvc._tcp.local, "QM" question
1148	37.112628	HP_46:1d:e8	Broadcast	ARP	60	Who has 172.16.15.136? Tell 172.16.15.17
1149	37.199409	HonHaiPrecis_f8:0e:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.99
1150	37.206860	HP_46:1a:5c	Broadcast	ARP	60	Who has 172.16.15.120? Tell 172.16.15.16
1151	37.324522	HP_46:15:ff	Broadcast	ARP	60	Who has 172.16.15.136? Tell 172.16.15.41

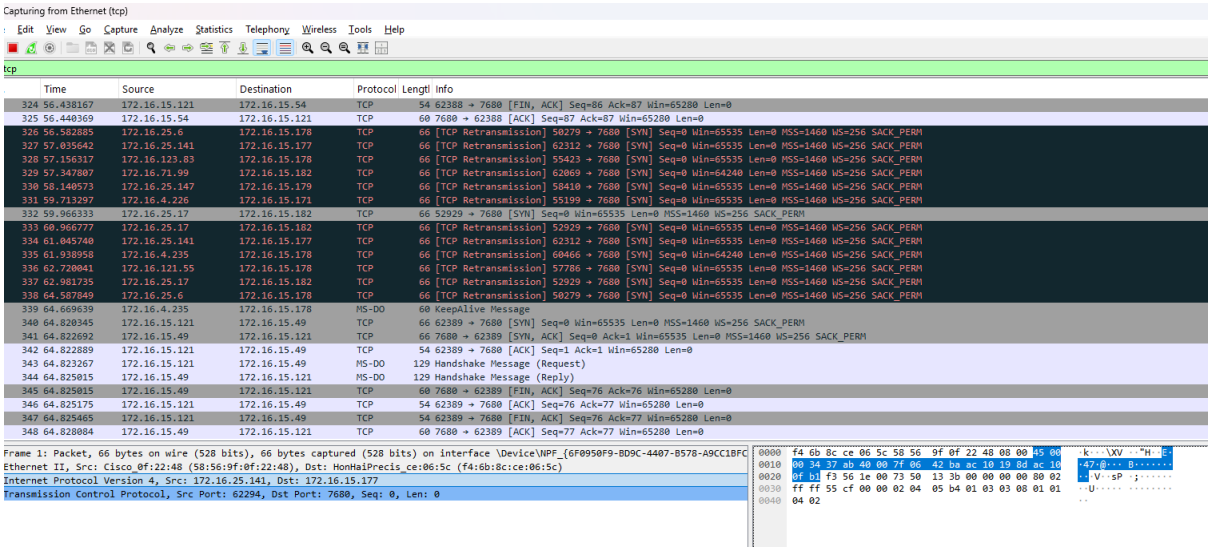
```
Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{6F0950F9-BD9C-4407-B578-A9CC1BFC
Ethernet II, Src: HonHaiPrecis_f6:d0:c3 (d0:f4:05:f6:d0:c3), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)
```

```
0000  ff ff ff ff ff ff d0 f4  05 f6 d0 c3 08 06 00 01   ........ ........
0010  08 00 06 04 00 01 d0 f4  05 f6 d0 c3 ac 10 0f 65   ........ .......e
0020  00 00 00 00 00 00 ac 10  0f 49 00 00 00 00 00 00   ........ .I......
0030  00 00 00 00 00 00 00 00  00 00 00 00               ........ ....
```

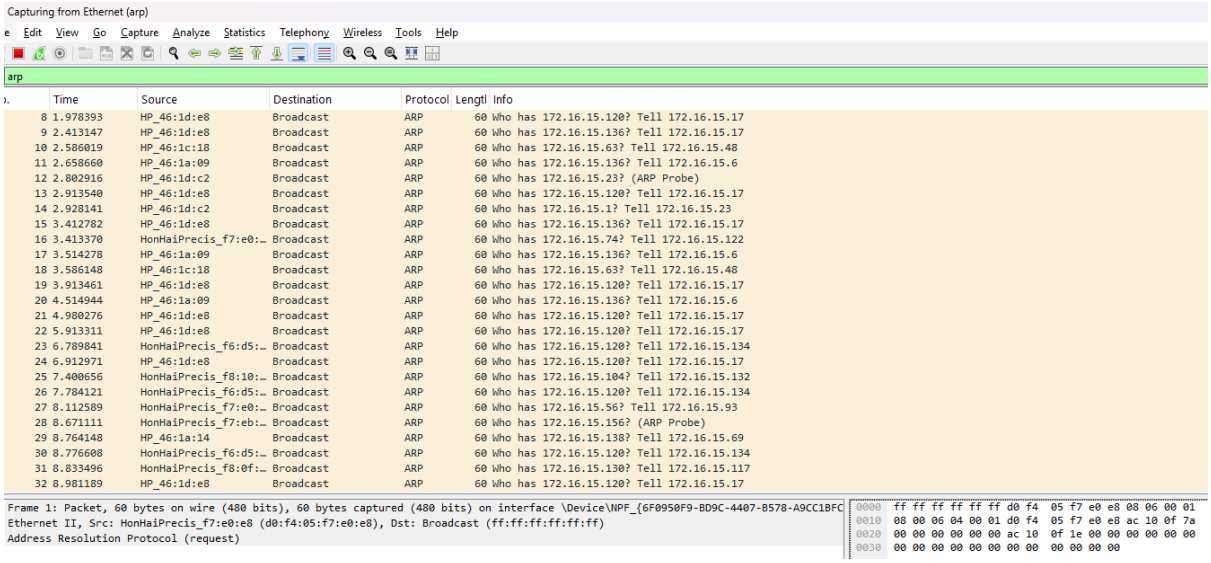
Using Capture and Display filter:



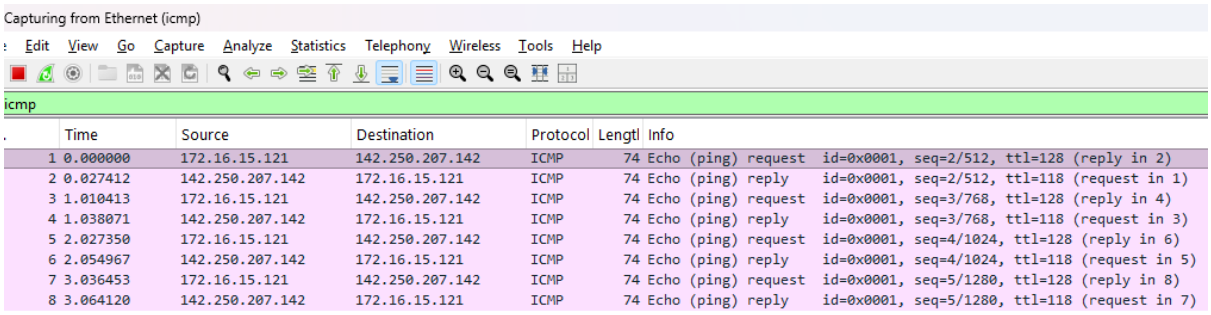
1.TCP packets isolated using display filter in Wireshark:



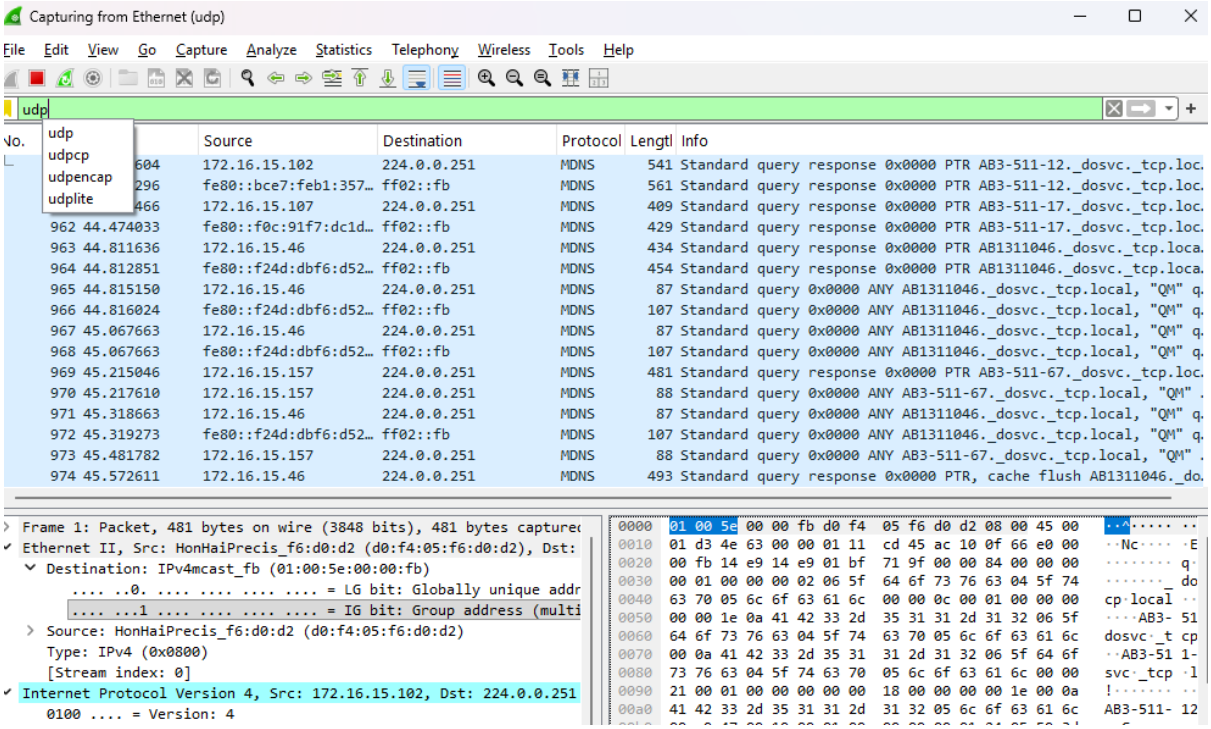
2.ARP packets isolated using display filter in Wireshark:



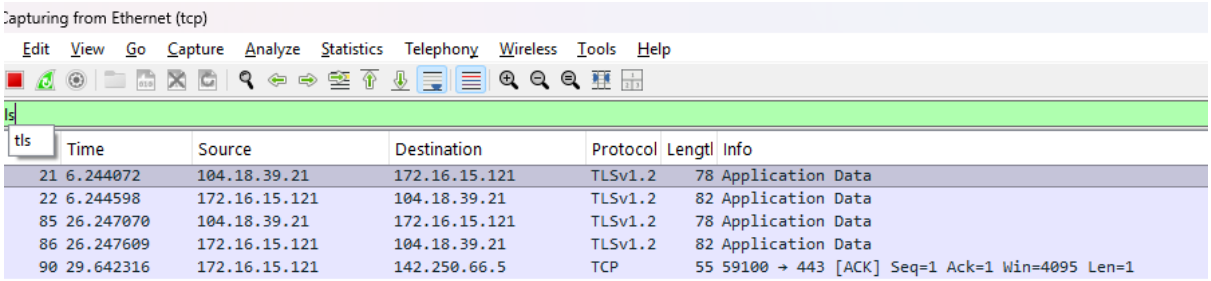
3.ICMP packets isolated using display filter in Wireshark:



4.UDP packets isolated using display filter in Wireshark:

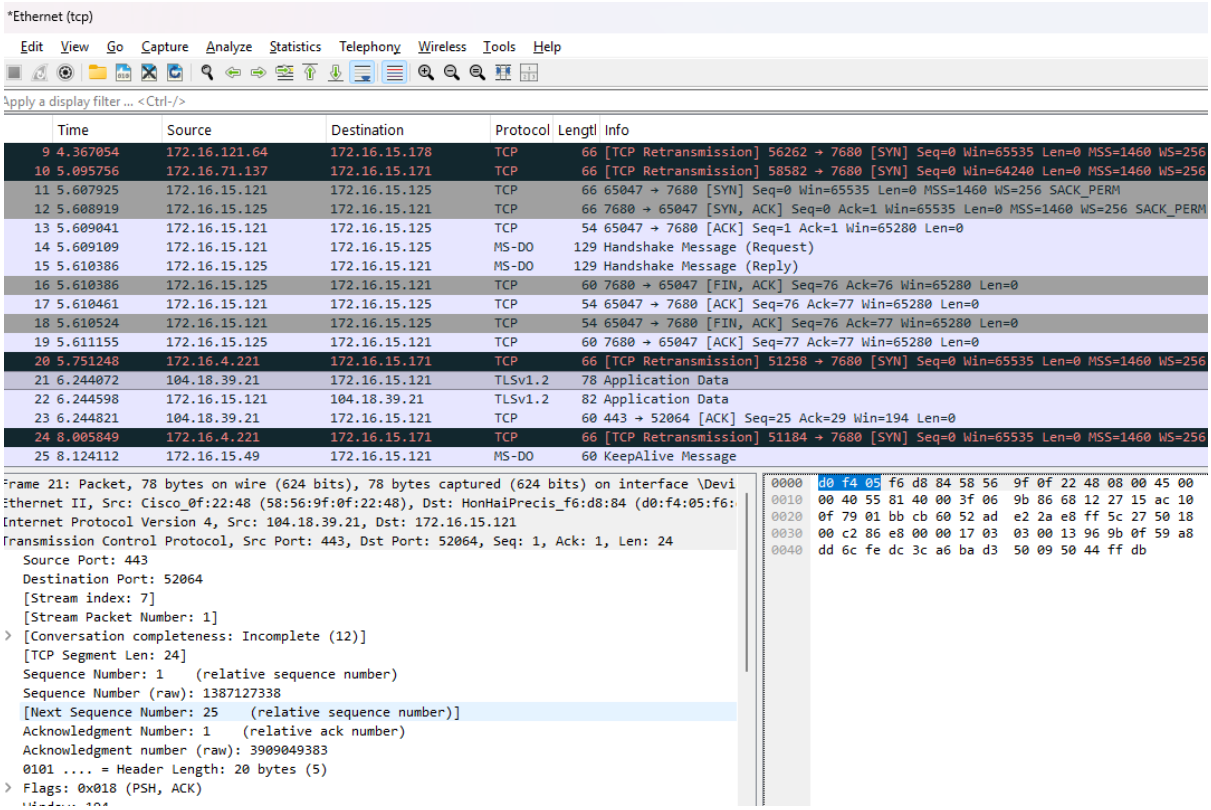


5.TLS packets isolated using display filter in Wireshark:



Capture any 5 screenshots from wireshark for all the 5 layers of TCP/ IP:

TRANSPORT LAYER:



PHYSICAL LAYER:

*Ethernet (tcp)

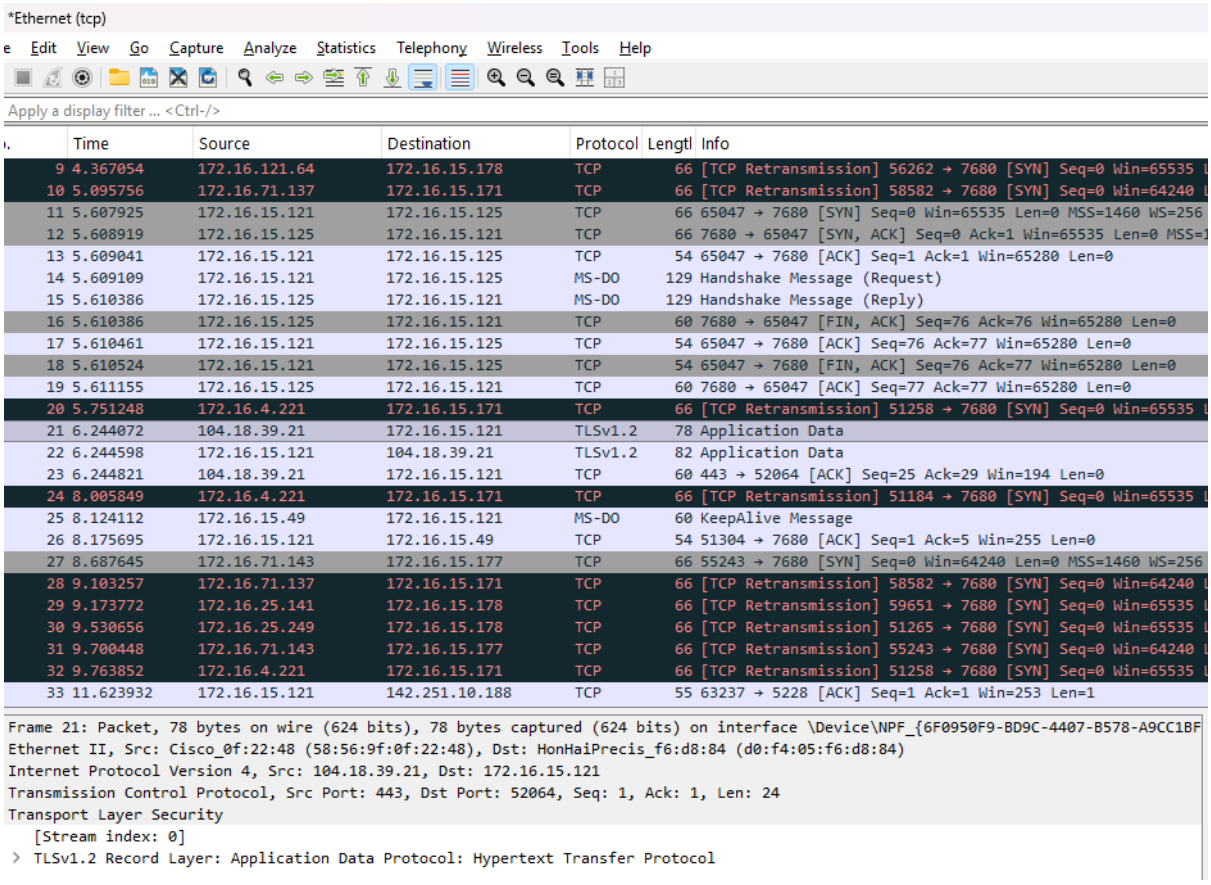
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Lengtl	Info
9	4.367054	172.16.121.64	172.16.15.178	TCP	66	[TCP Retransmission] 56262 → 7680 [SYN] Seq=0 Win=65535
10	5.095756	172.16.71.137	172.16.15.171	TCP	66	[TCP Retransmission] 58582 → 7680 [SYN] Seq=0 Win=64240
11	5.607925	172.16.15.121	172.16.15.125	TCP	66	65047 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256
12	5.608919	172.16.15.125	172.16.15.121	TCP	66	7680 → 65047 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=
13	5.609041	172.16.15.121	172.16.15.125	TCP	54	65047 → 7680 [ACK] Seq=1 Ack=1 Win=65280 Len=0
14	5.609109	172.16.15.121	172.16.15.125	MS-DO	129	Handshake Message (Request)
15	5.610386	172.16.15.125	172.16.15.121	MS-DO	129	Handshake Message (Reply)
16	5.610386	172.16.15.125	172.16.15.121	TCP	60	7680 → 65047 [FIN, ACK] Seq=76 Ack=76 Win=65280 Len=0
17	5.610461	172.16.15.121	172.16.15.125	TCP	54	65047 → 7680 [ACK] Seq=76 Ack=77 Win=65280 Len=0
18	5.610524	172.16.15.121	172.16.15.125	TCP	54	65047 → 7680 [FIN, ACK] Seq=76 Ack=77 Win=65280 Len=0
19	5.611155	172.16.15.125	172.16.15.121	TCP	60	7680 → 65047 [ACK] Seq=77 Ack=77 Win=65280 Len=0
20	5.751248	172.16.4.221	172.16.15.171	TCP	66	[TCP Retransmission] 51258 → 7680 [SYN] Seq=0 Win=65535
21	6.244072	104.18.39.21	172.16.15.121	TLSv1.2	78	Application Data
22	6.244598	172.16.15.121	104.18.39.21	TLSv1.2	82	Application Data
23	6.244821	104.18.39.21	172.16.15.121	TCP	60	443 → 52064 [ACK] Seq=25 Ack=29 Win=194 Len=0
24	8.005849	172.16.4.221	172.16.15.171	TCP	66	[TCP Retransmission] 51184 → 7680 [SYN] Seq=0 Win=65535
25	8.124112	172.16.15.49	172.16.15.121	MS-DO	60	KeepAlive Message
26	8.175695	172.16.15.121	172.16.15.49	TCP	54	51304 → 7680 [ACK] Seq=1 Ack=5 Win=255 Len=0
27	8.687645	172.16.71.143	172.16.15.177	TCP	66	55243 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256
28	9.103257	172.16.71.137	172.16.15.171	TCP	66	[TCP Retransmission] 58582 → 7680 [SYN] Seq=0 Win=64240
29	9.173772	172.16.25.141	172.16.15.178	TCP	66	[TCP Retransmission] 59651 → 7680 [SYN] Seq=0 Win=65535
30	9.530656	172.16.25.249	172.16.15.178	TCP	66	[TCP Retransmission] 51265 → 7680 [SYN] Seq=0 Win=65535
31	9.700448	172.16.71.143	172.16.15.177	TCP	66	[TCP Retransmission] 55243 → 7680 [SYN] Seq=0 Win=64240
32	9.763852	172.16.4.221	172.16.15.171	TCP	66	[TCP Retransmission] 51258 → 7680 [SYN] Seq=0 Win=65535
33	11.623932	172.16.15.121	142.251.10.188	TCP	55	63237 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1

```
Frame 21: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{6F0950F9-BD9C-4407-B578-A9CC1BF
    Section number: 1
  > Interface id: 0 (\Device\NPF_{6F0950F9-BD9C-4407-B578-A9CC1BFCB465})
    Encapsulation type: Ethernet (1)
    Arrival Time: Dec 19, 2025 14:41:46.773987000 India Standard Time
    UTC Arrival Time: Dec 19, 2025 09:11:46.773987000 UTC
    Epoch Arrival Time: 1766135506.773987000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 492.824000 milliseconds]
    [Time delta from previous displayed frame: 492.824000 milliseconds]
    [Time since reference or first frame: 6.244072000 seconds]
    Frame Number: 21
    Frame Length: 78 bytes (624 bits)
    Capture Length: 78 bytes (624 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:tls]
    Character encoding: ASCII (0)
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
```

APPLICATION LAYER:



*Ethernet (tcp)

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10	5.095756	172.16.71.137	172.16.15.171	TCP	66	[TCP Retransmission] 58582 → 7680 [SYN] Seq=0 Win=64240 L
11	5.607925	172.16.15.121	172.16.15.125	TCP	66	65047 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256
12	5.608919	172.16.15.125	172.16.15.121	TCP	66	7680 → 65047 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
13	5.609041	172.16.15.121	172.16.15.125	TCP	54	65047 → 7680 [ACK] Seq=1 Ack=1 Win=65280 Len=0
14	5.609109	172.16.15.121	172.16.15.125	MS-DO	129	Handshake Message (Request)
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18	5.610524	172.16.15.121	172.16.15.125	TCP	54	65047 → 7680 [FIN, ACK] Seq=76 Ack=77 Win=65280 Len=0
19	5.611155	172.16.15.125	172.16.15.121	TCP	60	7680 → 65047 [ACK] Seq=77 Ack=77 Win=65280 Len=0
20	5.751248	172.16.4.221	172.16.15.171	TCP	66	[TCP Retransmission] 51258 → 7680 [SYN] Seq=0 Win=65535 L
21	6.244072	104.18.39.21	172.16.15.121	TLSv1.2	78	Application Data
22	6.244598	172.16.15.121	104.18.39.21	TLSv1.2	82	Application Data
23	6.244821	104.18.39.21	172.16.15.121	TCP	60	443 → 52064 [ACK] Seq=25 Ack=29 Win=194 Len=0
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25	8.124112	172.16.15.49	172.16.15.121	MS-DO	60	KeepAlive Message
26	8.175695	172.16.15.121	172.16.15.49	TCP	54	51304 → 7680 [ACK] Seq=1 Ack=5 Win=255 Len=0
27	8.687645	172.16.71.143	172.16.15.177	TCP	66	55243 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256
28	9.103257	172.16.71.137	172.16.15.171	TCP	66	[TCP Retransmission] 58582 → 7680 [SYN] Seq=0 Win=64240 L
29	9.173772	172.16.25.141	172.16.15.178	TCP	66	[TCP Retransmission] 59651 → 7680 [SYN] Seq=0 Win=65535 L
30	9.530656	172.16.25.249	172.16.15.178	TCP	66	[TCP Retransmission] 51265 → 7680 [SYN] Seq=0 Win=65535 L
31	9.700448	172.16.71.143	172.16.15.177	TCP	66	[TCP Retransmission] 55243 → 7680 [SYN] Seq=0 Win=64240 L
32	9.763852	172.16.4.221	172.16.15.171	TCP	66	[TCP Retransmission] 51258 → 7680 [SYN] Seq=0 Win=65535 L
33	11.623932	172.16.15.121	142.251.10.188	TCP	55	63237 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1

```
Frame 21: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{6F0950F9-BD9C-4407-B578-A9CC1BF
Ethernet II, Src: Cisco_0f:22:48 (58:56:9f:0f:22:48), Dst: HonHaiPrecis_f6:d8:84 (d0:f4:05:f6:d8:84)
Internet Protocol Version 4, Src: 104.18.39.21, Dst: 172.16.15.121
Transmission Control Protocol, Src Port: 443, Dst Port: 52064, Seq: 1, Ack: 1, Len: 24
Transport Layer Security
   [Stream index: 0]
> TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
```

DATA LINK LAYER:

No.	Time	Source	Destination	Protocol	Length	Info
17	3.411365	HP_46:1a:48	Broadcast	ARP	60	Who has 172.16.15.23? Tell 172.16.15.21
18	3.624273	HonHaiPrecis_f6:d0:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.109
19	3.702925	HP_45:db:2e	Broadcast	ARP	60	Who has 172.16.15.23? Tell 172.16.15.37
20	3.702925	HP_45:db:2e	Broadcast	ARP	60	Who has 172.16.15.63? Tell 172.16.15.37
21	3.702925	HP_45:db:2e	Broadcast	ARP	60	Who has 172.16.15.4? Tell 172.16.15.37
22	3.709801	HonHaiPrecis_f8:0f:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.143
23	3.886898	HP_45:db:1b	Broadcast	ARP	60	Who has 172.16.15.120? Tell 172.16.15.30
24	3.989694	HonHaiPrecis_f7:de:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.146
25	3.989694	HonHaiPrecis_f7:de:…	Broadcast	ARP	60	Who has 172.16.15.138? Tell 172.16.15.146
26	4.411205	HP_46:1a:48	Broadcast	ARP	60	Who has 172.16.15.23? Tell 172.16.15.21
27	4.508627	HonHaiPrecis_3e:65:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.100
28	4.593513	HonHaiPrecis_f7:e8:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.139
29	4.705240	HP_45:db:2e	Broadcast	ARP	60	Who has 172.16.15.23? Tell 172.16.15.37
30	4.705240	HP_45:db:2e	Broadcast	ARP	60	Who has 172.16.15.63? Tell 172.16.15.37
31	4.705240	HP_45:db:2e	Broadcast	ARP	60	Who has 172.16.15.4? Tell 172.16.15.37
32	4.708693	HonHaiPrecis_f8:0f:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.143
33	4.880391	HP_45:db:1b	Broadcast	ARP	60	Who has 172.16.15.120? Tell 172.16.15.30
34	5.184180	HonHaiPrecis_3e:65:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.100
35	5.331038	HonHaiPrecis_f7:e8:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.139
36	5.891017	HP_45:db:1b	Broadcast	ARP	60	Who has 172.16.15.120? Tell 172.16.15.30
37	5.999991	HP_46:15:ff	Broadcast	ARP	60	Who has 172.16.15.136? Tell 172.16.15.41
38	6.195191	HonHaiPrecis_3e:65:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.100
39	6.327356	HonHaiPrecis_f7:e8:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.139
40	6.375628	HonHaiPrecis_f6:d0:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.109
41	6.507909	HonHaiPrecis_f8:0f:…	Broadcast	ARP	60	Who has 172.16.15.73? Tell 172.16.15.145

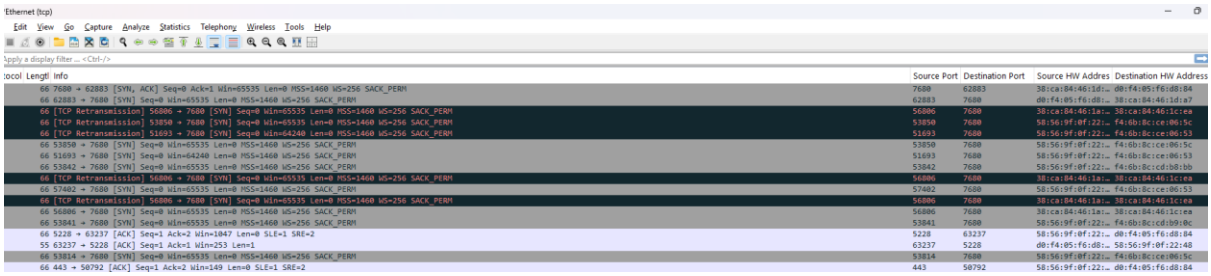
```
> Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{6F0950F9-BD9C-4407-B578-A9CC1B
v Ethernet II, Src: HP_46:1c:b2 (38:ca:84:46:1c:b2), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  v Destination: Broadcast (ff:ff:ff:ff:ff:ff)
      .... ..1. .... .... .... .... = LG bit: Locally administered address (this is NOT the factory default)
      .... ...1 .... .... .... .... = IG bit: Group address (multicast/broadcast)
  v Source: HP_46:1c:b2 (38:ca:84:46:1c:b2)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: ARP (0x0806)
    [Stream index: 0]
    Padding: 000000000000000000000000000000000000
v Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: HP_46:1c:b2 (38:ca:84:46:1c:b2)
    Sender IP address: 172.16.15.15
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 172.16.15.73
```

NETWORK LAYER:

	Time	Source	Destination	Protocol	Lengtl	Info
1	0.000000	172.16.25.141	172.16.15.178	TCP	66	61667 → 7680 [SYN] S
2	3.639117	172.64.148.235	172.16.15.121	TLSv1.2	78	Application Data
3	3.639534	172.16.15.121	172.64.148.235	TLSv1.2	82	Application Data
4	3.639794	172.64.148.235	172.16.15.121	TCP	60	443 → 50003 [ACK] Se
5	3.935495	172.16.15.121	172.16.15.140	TCP	66	62877 → 7680 [SYN] S
6	3.936671	172.16.15.140	172.16.15.121	TCP	66	7680 → 62877 [SYN, A
7	3.936811	172.16.15.121	172.16.15.140	TCP	54	62877 → 7680 [ACK] S
8	3.937139	172.16.15.121	172.16.15.140	MS-DO	129	Handshake Message (R
9	3.937914	172.16.15.140	172.16.15.121	MS-DO	129	Handshake Message (R
10	3.938332	172.16.15.121	172.16.15.140	MS-DO	64	BitField Message (ha
11	3.943317	172.16.15.140	172.16.15.121	MS-DO	64	BitField Message (ha
12	3.943317	172.16.15.140	172.16.15.121	TCP	60	7680 → 62877 [FIN, A
13	3.943409	172.16.15.121	172.16.15.140	TCP	54	62877 → 7680 [ACK] S
14	3.943855	172.16.15.121	172.16.15.140	TCP	54	62877 → 7680 [FIN, A
15	3.944241	172.16.15.140	172.16.15.121	TCP	60	7680 → 62877 [ACK] S
16	8.014802	172.16.25.141	172.16.15.178	TCP	66	[TCP Retransmission]

```
Frame 1: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\N
Ethernet II, Src: Cisco_0f:22:48 (58:56:9f:0f:22:48), Dst: HonHaiPrecis_ce:06:53 (f4:6b:8c:ce:06:
Internet Protocol Version 4, Src: 172.16.25.141, Dst: 172.16.15.178
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 52
    Identification: 0x13de (5086)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 127
    Protocol: TCP (6)
    Header Checksum: 0x6686 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.16.25.141
    Destination Address: 172.16.15.178
    [Stream index: 0]
```

Add the 4 columns called source port, destination port, source hw, address and destination hw address and capture the screenshot of wireshark home page with added columns.



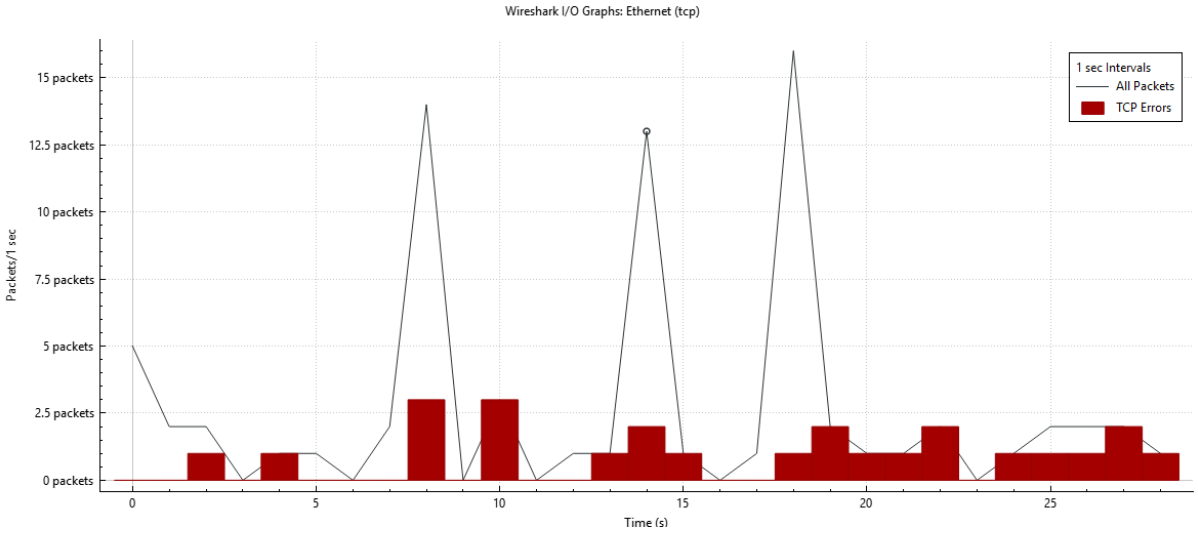
CONVERSATIONS FROM STATISTICS MENU:

Ethernet (tcp)

Ethernet · 12 | IPv4 · 18 | IPv6 | TCP · 18 | UDP

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
38:ca:84:46:1a:51	38:ca:84:46:1c:ea	4	264 bytes	3	4	264 bytes	0	0 bytes	1.490798	7.0130	301 bits/s	0 bits/s
58:56:9f:0f:22:48	f4:6b:8c:cd:b8:b5	3	198 bytes	11	3	198 bytes	0	0 bytes	25.252967	3.0302	522 bits/s	0 bits/s
58:56:9f:0f:22:48	f4:6b:8c:cd:b8:bb	1	66 bytes	5	1	66 bytes	0	0 bytes	5.421942	0.0000		
58:56:9f:0f:22:48	f4:6b:8c:cd:b9:0c	1	66 bytes	2	1	66 bytes	0	0 bytes	1.350333	0.0000		
58:56:9f:0f:22:48	f4:6b:8c:ce:05:e9	5	330 bytes	7	5	330 bytes	0	0 bytes	12.894800	15.0267	175 bits/s	0 bits/s
58:56:9f:0f:22:48	f4:6b:8c:ce:06:53	7	462 bytes	4	7	462 bytes	0	0 bytes	2.615382	19.7663	186 bits/s	0 bits/s
58:56:9f:0f:22:48	f4:6b:8c:ce:06:5c	12	792 bytes	1	12	792 bytes	0	0 bytes	0.217276	27.2339	232 bits/s	0 bits/s
58:56:9f:0f:22:48	f4:6b:8c:ce:08:a4	4	264 bytes	9	4	264 bytes	0	0 bytes	18.671142	7.0364	300 bits/s	0 bits/s
d0:f4:05:f6:d8:84	38:ca:84:46:1c:6c	11	836 bytes	8	6	439 bytes	5	397 bytes	14.578806	0.0159	220 kbps	199 kbps
d0:f4:05:f6:d8:84	38:ca:84:46:1d:a7	11	829 bytes	6	6	454 bytes	5	375 bytes	8.553104	0.0103	354 kbps	292 kbps
d0:f4:05:f6:d8:84	58:56:9f:0f:22:48	7	462 bytes	0	3	192 bytes	4	270 bytes	0.000000	18.6475	82 bits/s	115 bits/s
d0:f4:05:f6:d8:84	d0:f4:05:f8:10:42	11	800 bytes	10	6	421 bytes	5	379 bytes	18.954791	0.0121	278 kbps	250 kbps

IO GRAPHS:



SCAPY OUTPUT:

```
PS C:\Users\Student\24BCE1389> python scapy_analysis.py
Total packets: 77
Number of Source IPs: 16
Number of Destination IPs: 15
Number of Protocols Used: 1
Packets per protocol:
6 : 77
```

CODE:

```
from scapy.all import rdpcap, IP

pkts = rdpcap(r"C:\Users\Student\24BCE1389\exp3.pcap")

print("Total packets:", len(pkts))

src_ips = set()
dst_ips = set()
protocols = {}

for pkt in pkts:
    if IP in pkt:
        src_ips.add(pkt[IP].src)
        dst_ips.add(pkt[IP].dst)
        proto = pkt[IP].proto
        protocols[proto] = protocols.get(proto, 0) + 1

print("Number of Source IPs:", len(src_ips))
print("Number of Destination IPs:", len(dst_ips))
print("Number of Protocols Used:", len(protocols))

print("Packets per protocol:")
for proto, count in protocols.items():
    print(proto, ":", count)
```